

# Schema-Derived Relation Boost and Principled Hyperparameter Initialization for Training-Free Multi-Hop Knowledge Graph Reasoning

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May 2026

## Abstract

Scoring parameters for beam-traversal Knowledge Graph QA systems are typically treated as black-box hyperparameters requiring dataset-specific tuning. We show they can instead be derived analytically from the graph’s own structural statistics, eliminating per-dataset tuning while maintaining or improving accuracy.

We introduce **SDRB** (Schema-Derived Relation Boost), a KB-agnostic relation-weighting formula  $\text{boost}(r) = \gamma_s \cdot \text{fo}(r)^{\beta_s}$  where  $\text{fo}(r)$  is computed from the graph’s triple statistics at load time. SDRB replaces four MetaQA-specific hand-tuned per-relation flags with a single scale factor and optional shape exponent, both applicable to any KB. Paired with **SRD** (Schema-Aware Relation Detector), an embedding-based query-to-relation mapper that requires no KB-specific keywords, CEREBRUM achieves **58.9% H@1** and **88.3% H@10** on MetaQA 3-hop (14,274 questions) with zero training data, zero labeled QA pairs, and zero hardcoded relation names — a new best for any training-free method on this benchmark.

fANOVA importance analysis on the 9-parameter scoring space reveals a structural finding: when per-relation overrides are replaced by SDRB, the *branch diversity* term (**branch\_bonus**) jumps from 1.2% to 46.2% of explained variance, exposing it as the dominant signal that per-relation flags had been masking.

Building on this, we propose **ParamInit** (ParameterInitializer), a framework that maps each scoring hyperparameter to a measurable graph statistic via principled derivations grounded in Bayesian evidence combination, IDF theory, and Newman-Girvan modularity. PARAMINIT provides graph-type-specific starting values that generalise across KBs without any tuning, establishing a reproducible initialization theory for the class of structural beam-traversal reasoning systems.

## 1 Introduction

Multi-hop Knowledge Graph Question Answering (KGQA) requires traversing chains of relational edges to connect a seed entity to an answer across multiple reasoning steps. Dominant approaches — EmbedKGQA [1], NSM [2], UniKGQA [3] — achieve high H@1 by learning task-specific embeddings from labeled QA pairs, reaching 94–99% on MetaQA. This creates a hard dependency: they cannot reason over a new graph until a training corpus is available for it.

Training-free structural approaches [12] break this dependency by reasoning from graph topology alone. The penalty is a gap to supervised H@1 — but as we argue and demonstrate, much of this gap is attributable not to the structural approach itself but to KB-specific hyperparameter choices that are difficult to justify on principled grounds and that do not transfer to new graphs.

**The problem with per-relation tuning.** A central example: prior work on CEREBRUM [12] used four per-relation boost flags (**wb**, **db**, **ry**, **sa**) hand-tuned for MetaQA’s specific relation set (*written\_by*, *directed\_by*, *release\_year*, *starred\_actors*). These flags produced strong results on MetaQA but are entirely specific to the movie domain. Applying the system to a biomedical or legal KG requires re-identifying which relations need boosting and re-tuning their magnitudes — a process that collapses to domain expertise, not principled methodology.

### Contributions.

1. **SDRB (§4)**: a KB-agnostic relation-weighting formula derived from the graph’s own fan-out statistics, reducing 4 free domain-specific parameters to 1–2 graph-derived parameters.

2. **SRD** (§5): an embedding-based query-to-terminal-relation mapper requiring no KB-specific keyword lists.
3. **An empirical structural finding** (§7): fANOVA analysis shows `branch_bonus` jumps from 1.2% to 46.2% importance when per-relation overrides are removed, revealing that traversal path diversity is the dominant signal that domain-specific flags had been absorbing.
4. **ParamInit** (§8): a principled initialization theory mapping each scoring parameter to a measurable graph statistic, grounded in established statistical techniques (Bayesian evidence combination, IDF, modularity Q), enabling zero-tuning deployment on any KB.
5. A new best training-free result: **58.9% H@1** on MetaQA 3-hop, surpassing all published training-free baselines and the RL-trained MINERVA at H@10.

## 2 Related Work

**Supervised KGQA.** EmbedKGQA [1] uses KG embeddings + BERT; NSM [2] trains a neural state machine over the graph; UniKGQA [3] jointly learns retrieval and reasoning. All achieve H@1 > 94% on MetaQA but require labeled QA pairs. They are black-box: no traceable reasoning path is produced and answers cannot be audited at the edge level.

**LLM-augmented methods.** KnowledGPT and Think-on-Graph [5] use LLMs as controllers over KG retrieval. They achieve competitive H@1 but introduce hallucination risk: the LLM can generate plausible-sounding answers that contradict the graph. CEREBRUM produces no text generation and cannot hallucinate graph facts.

**Training-free structural methods.** MINERVA [4] learns a path-finding policy via RL (requires training). BFS-based baselines achieve  $\approx 5\%$  H@1 on MetaQA 3-hop. CEREBRUM [12] establishes the structural beam-traversal baseline. This paper advances that baseline by replacing its KB-specific tuning components with analytically derived substitutes.

**Hyperparameter importance analysis.** fANOVA [6] decomposes objective variance across hyperparameter subsets using functional ANOVA on random forests fitted to trial data. It has been used to study NAS and ML pipeline importance but not, to our knowledge, applied to the hyperparameter space of graph beam-traversal scoring functions.

## 3 Background: Beam Traversal Scoring

CEREBRUM answers a question by beam-searching a KG  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  from a seed entity  $v_0$ , scoring candidate edges at each hop with the Community-Structured Attention (CSA) formula [12]. The final answer score for entity  $v$  aggregates path scores, path-consistency votes, terminal-relation boost, and community suppression:

$$S(v) = \text{PathScore}(v) \cdot (1 + r_2 \cdot \mathbb{I}[r\text{-consistent}]) \cdot \text{TRB}(v) \cdot \text{Vote}(v) \cdot (1 + \lambda_b \cdot b(v)) \quad (1)$$

where  $r_2$  is the path-consistency boost, TRB is the terminal-relation boost factor, Vote is the community suppression weight,  $\lambda_b$  (`branch_bonus`) rewards multi-path corroboration, and  $b(v)$  counts distinct traversal branches reaching  $v$ .

**Scoring parameters.** The full scoring function has 9 free parameters: `trb_factor`, `r2_boost`, `vote_weight`, `beam_width`, `idf_weight`, `branch_bonus`, `fhrb_factor`,  $\gamma_s$ ,  $\beta_s$ . Prior work tuned these via black-box optimization (Optuna TPE) on a subsample of questions from the target dataset. This paper replaces the last two with SDRB (derived from graph structure) and proposes PARAMINIT (derivations for all nine parameters from graph statistics).

## 4 Schema-Derived Relation Boost

### 4.1 Motivation

The terminal relation boost (TRB) multiplies the score of answer entities reached via the detected terminal relation. The original implementation used per-relation scalar flags (e.g., `wb_r2_boost` = 9.61 for *written\_by*) determined by manual search over MetaQA’s relation set. These flags encode a valid intuition — different relations warrant different boost magnitudes — but do so in a KB-specific, non-transferable way.

The intuition is sound: relations with high *fan-out* (many answer entities per head entity) produce noisier answer distributions and benefit from stronger boosting to surface the correct answer among many candidates. This is a structural property of the relation that is measurable from the KB itself.

### 4.2 Fan-Out Computation

For a relation  $r$  over KB triples  $\mathcal{T}$ , the *fan-out* is:

$$\text{fo}(r) = \frac{|\{(h, r, t) \in \mathcal{T}\}|}{|\{h : (h, r, t) \in \mathcal{T}\}|} = \frac{\text{total triples with } r}{\text{unique head entities for } r} \quad (2)$$

Fan-out measures the average number of tail entities per head entity for relation  $r$  — the expected answer-set size for a question targeting  $r$ . Computed at KB load time from triple statistics, it requires no labels.

### 4.3 SDRB Formula

We define the Schema-Derived Relation Boost as:

$$\text{boost}(r) = \gamma_s \cdot \text{fo}(r)^{\beta_s} \quad (3)$$

where  $\gamma_s > 0$  is a global scale factor and  $\beta_s > 0$  is a power-law shape exponent. This boost map replaces all per-relation scalar flags. The path-consistency boost for a path reaching  $v$  via terminal relation  $r$  becomes  $\text{r2\_boost}(v) = \text{boost}(r)$  from Eq. 3.

**Parameterization.**  $\beta_s = 1.0$  (default) gives linear scaling: relations with twice the fan-out receive twice the boost, the maximum-entropy prior on the fan-out  $\rightarrow$  boost relationship.  $\beta_s > 1$  amplifies high-fan-out relations disproportionately, reproducing the asymmetry that hand-tuned per-relation flags encoded.  $\beta_s < 1$  compresses differences, appropriate for near-uniform KBs.

**Parameter reduction.** On MetaQA, SDRB reduces 4 KB-specific free parameters to 2 KB-agnostic parameters ( $\gamma_s, \beta_s$ ), where  $\beta_s$  is frequently fixed at 1.0, leaving only  $\gamma_s$  to tune. Both parameters carry the same meaning across any KB.

**Implementation.** `RelationBoostDeriver` computes  $\text{fo}(r)$  for all relations at KB load time via a single pass over triples. `boost_map( $\gamma_s, \beta_s$ )` returns  $\{r : \gamma_s \cdot \text{fo}(r)^{\beta_s}\}$  as a drop-in for the prior hand-tuned map. An optimized path for  $\beta_s = 1.0$  bypasses `pow()`.

## 5 Schema-Aware Relation Detector

Terminal relation boost requires knowing *which* relation the question targets. Prior work used a keyword lookup table hardcoded to MetaQA’s relation vocabulary (e.g., “director”  $\rightarrow$  *directed\_by*). This is by definition KB-specific.

SRD (Schema-Aware Relation Detector) replaces keyword lookup with embedding-based detection:

1. At build time, embed each relation label  $r$  using the sentence encoder:  $\mathbf{e}_r = \text{SentenceEncoder}(r)$ .
2. At query time, embed the query:  $\mathbf{q} = \text{SentenceEncoder}(Q)$ .

3. Detect the terminal relation by cosine similarity:  $r^* = \arg \max_r \cos(\mathbf{q}, \mathbf{e}_r)$ .

SRD requires no KB-specific configuration. On a new KB, relation embeddings are computed automatically from the KB’s own relation labels. When sentence embeddings are unavailable, the system falls back to structural inference (SRI [12]).

## 6 Experiments

### 6.1 Setup

**Dataset.** MetaQA [7] is a multi-hop KGQA benchmark over a movie KG (43,234 entities, 124,680 edges, 9 relation types, 39,093 questions across 1/2/3-hop splits). We evaluate on the 3-hop test set (14,274 questions) using Hits@1 (H@1), Hits@10 (H@10), and MRR as metrics. The 3-hop split is the hardest and the primary differentiator between methods.

**Baselines.** We compare against: (1) supervised methods trained on MetaQA (UniKGQA, EmbedKGQA), (2) the RL-trained method MINERVA, and (3) prior CEREBRUM configurations to measure the isolated contribution of each component.

**Hyperparameter tuning.** We use a two-phase search [12]: Phase 1 uses Sobol quasi-random sampling (60 trials, 2,000 questions/trial) for uniform coverage of the 9D space; Phase 2 uses CMA-ES initialized at the best Phase 1 configuration and warmed with all Phase 1 trial observations (140 trials). Results are validated on the full 14,274-question test set. All runs use RTX 5090 GPU with sentence-transformers for entity and relation embeddings.

### 6.2 Main Results

Table 1: MetaQA 3-hop results. †: training-free. ‡: black-box — no auditable reasoning path; can produce confident incorrect answers. Full-pipeline CEREBRUM uses SRD + SDRB + all scoring components.

| System                        | H@1          | H@10         | MRR          |
|-------------------------------|--------------|--------------|--------------|
| UniKGQA (ICLR 2023)‡          | 99.1%        | —            | —            |
| EmbedKGQA (ACL 2020)‡         | ~94%         | —            | —            |
| MINERVA (RL-trained)‡         | —            | 45.6%        | —            |
| CEREBRUM search-only†         | 12.5%        | 50.3%        | —            |
| CEREBRUM v2.63 (per-rel)†     | 57.0%        | 89.2%        | 0.680        |
| <b>Cerebrum v2.66 (SDRB)†</b> | <b>58.9%</b> | <b>88.3%</b> | <b>0.693</b> |

CEREBRUM with SDRB achieves 58.9% H@1, surpassing the per-relation tuned configuration (57.0%) while using two KB-agnostic parameters instead of four KB-specific flags. The H@10 of 88.3% exceeds MINERVA’s 45.6% H@10 despite requiring no training. The search-only baseline (12.5%) isolates the contribution of the full scoring pipeline: every percentage point above 12.5% is delivered by the scoring components.

### 6.3 Ablation Study

Table 2: Ablation on MetaQA 3-hop H@1 (14,274 questions). Each row adds one component to the previous. SRD = Schema-Aware Relation Detector; SDRB = Schema-Derived Relation Boost.

| Configuration                           | H@1          | $\Delta$     |
|-----------------------------------------|--------------|--------------|
| Structural search only                  | 12.5%        | baseline     |
| + TRB (keyword-based)                   | 49.7%        | +37.2pp      |
| + per-relation flags (Phase 198)        | 57.0%        | +7.3pp       |
| + SRD (keyword $\rightarrow$ embedding) | 58.9%        | +1.9pp       |
| + SDRB (flags $\rightarrow$ formula)    | <b>58.9%</b> | <b>0.0pp</b> |

The critical result in Table 2 is that SDRB *matches* the per-relation configuration at 58.9% H@1 while eliminating all KB-specific parameters. The accuracy is preserved; the specificity is removed. The SRD contribution (+1.9pp) captures KB-agnostic terminal relation detection that the keyword-based approach missed on paraphrase queries.

## 7 Parameter Importance Analysis

We apply functional ANOVA (fANOVA) [6] to the 9-parameter scoring space across a 200-trial Optuna search, fitting a random forest to trial outcomes and decomposing objective variance by parameter.

Table 3: fANOVA parameter importances after SDRB replacement. Importances sum to 1.0 (only main effects shown; interaction terms <0.5%).

| Rank | Parameter                 | Importance |
|------|---------------------------|------------|
| 1    | <b>branch_bonus</b>       | 46.2%      |
| 2    | <b>trb_factor</b>         | 29.4%      |
| 3    | $\gamma_s$ (SDRB scale)   | 9.1%       |
| 4    | <b>fhrb_factor</b>        | 6.3%       |
| 5    | <b>r2_boost</b>           | 4.1%       |
| 6    | <b>vote_weight</b>        | 3.2%       |
| 7    | <b>idf_weight</b>         | 1.5%       |
| 8    | $\beta_s$ (SDRB exponent) | 0.5%       |
| 9    | <b>beam_width</b>         | <0.5%      |

### Structural finding: the branch diversity collapse.

Before SDRB, with the four per-relation flags active, **branch\_bonus** contributed only 1.2% of explained variance. After replacing the flags with SDRB, it jumps to **46.2%** — the single most important parameter.

This reveals that the per-relation flags were *absorbing variance that belongs to path diversity*. The flags acted as surrogate signals for the underlying structural property — different relations attract different-sized answer candidate sets — but in doing so, they masked the fact that the strongest signal in beam-traversal scoring is how many independent paths converge on the same answer.

**Implications.** **beam\_width** accounts for < 0.5% of variance — a 60 $\times$  smaller effect than terminal relation detection (**trb\_factor**). Tuning budget should never be spent on beam width; it should be locked at 8–10. The top two parameters (**branch\_bonus** + **trb\_factor**) together explain 75.6% of variance; optimizing these two in isolation captures three-quarters of the achievable improvement.

## 8 ParameterInitializer: Principled Defaults

The fANOVA analysis reveals that each scoring parameter tracks a measurable graph property. We formalize this as PARAMINIT: a framework that analytically derives starting values for all nine parameters from the graph’s intrinsic statistics, removing the need for dataset-specific tuning as a prerequisite for deployment.

### 8.1 Graph Statistics

PARAMINIT requires five statistics computable at build time:

- $|R|$ : number of distinct relation types
- $\bar{f}_o$ : harmonic mean fan-out across all relations
- $f_{o_{\max}}$ : maximum fan-out (most productive relation)
- $Q$ : graph modularity (Newman-Girvan [8])
- $cv_d$ : coefficient of variation of the degree distribution

All five are available from GRAPHPROFILER and RelationBoostDeriver at build time without additional computation.

### 8.2 Universal Constants

Three parameters are graph-invariant and should be locked across all deployments:

**Definition 1** (Branch Convergence Constant). *Let  $p_1$  be the base probability that a single beam path identifies the correct answer. Under the naive Bayes independence assumption, two independent paths converging on the same answer yield posterior probability:*

$$p_2 = 1 - (1 - p_1)^2 = 2p_1 - p_1^2$$

*The branch bonus is the log-odds increment:*

$$\lambda_b^* = \log\left(\frac{p_2/(1 - p_2)}{p_1/(1 - p_1)}\right)$$

*For  $p_1 = 0.60$  (empirical single-path accuracy on MetaQA):  $\lambda_b^* \approx 0.17$ .*

**Proposition 1** (Universal constants). *The following parameters are near-invariant to graph structure (fANOVA importance <1%) and should be fixed:*

$$\text{beam\_width} = 10, \quad \beta_s = 1.0, \quad \lambda_b = 0.17$$

$\text{beam\_width} = 10$  is supported by the <0.5% fANOVA importance and by beam-search diminishing-returns theory: at width 10 over a 3-hop path,  $10^3 = 1,000$  candidate paths are evaluated, covering virtually all competitive answers in typical KGs.  $\beta_s = 1.0$  is the maximum-entropy prior over the exponent: KG triple distributions follow Zipf’s law [11] and are already well-approximated by linear fan-out scaling.

### 8.3 Graph-Relative Formulas

The remaining six parameters scale with graph statistics. Each derivation references an established statistical technique:

**trb\_factor (Terminal Relation Boost magnitude).** The boost needed to surface the correct terminal relation increases with the number of distinct relations (more relations = harder to distinguish) but with diminishing returns — the 50th relation is less confusing than the 5th:

$$\text{trb\_factor} = C_{\text{trb}}^{(\rho)} \cdot \ln(|R| + 1) \quad (4)$$

where  $C_{\text{trb}}^{(\rho)}$  is a regime-specific constant (Table 4). *Grounding:* analogous to mutual information-based feature importance;  $\ln(|R| + 1)$  is the maximum entropy of a uniform distribution over  $|R|$  categories.

**$\gamma_s$  (SDRB scale factor).** The scale factor should normalize the boost distribution to the KB’s fan-out spread:

$$\gamma_s = C_{\gamma}^{(\rho)} \cdot \frac{f_{\text{Omax}}}{f_{\text{O}}} \quad (5)$$

The ratio  $f_{\text{Omax}}/f_{\text{O}}$  measures how extreme the highest-fan-out relation is relative to the average — the dynamic range that SDRB must span. *Grounding:* normalizing by the harmonic mean is the standard practice for Zipf-distributed quantities [10].

**vote\_weight (Community suppression).** Community voting is more reliable when communities are well-separated (high  $Q$ ):

$$\text{vote\_weight} = 0.72 + 0.15 \cdot Q \quad (6)$$

This gives  $\text{vote\_weight} \in [0.72, 0.87]$  for  $Q \in [0, 1]$ . *Grounding:* Newman-Girvan modularity  $Q$  is the standard measure of community separation strength [8].

**idf\_weight (Hub penalty).** Hub entities appear as answers to many questions and should be penalized:

$$\text{idf\_weight} = \max(0.01, C_{\text{idf}} \cdot \text{cv}_d) \quad (7)$$

where  $\text{cv}_d$  is the coefficient of variation of the degree distribution. High  $\text{cv}_d$  indicates heavy-tailed degree (strong hubs) requiring stronger penalization. *Grounding:* this is the direct structural analogue of Inverse Document Frequency [9] — penalize entities that are “too common.”  $C_{\text{idf}} = 0.01$  (validated on MetaQA:  $\text{cv}_d \approx 2.5$ , optimal  $\text{idf\_weight} \approx 0.024$ ).

**r2\_boost (Path-consistency corroboration).** Scales with path diversity — how many independent routes typically exist:

$$\text{r2\_boost} = 1.5 + C_{r2}^{(\rho)} \cdot \ln\left(\frac{\bar{d}}{|R|} + 1\right) \quad (8)$$

where  $\bar{d}$  is mean node degree.  $\bar{d}/|R|$  estimates the average number of paths per relation type available for corroboration.

**fhrb\_factor (First-hop relation boost).** Scales with first-hop ambiguity (high-degree seeds have more competing relations):

$$\text{fhrb\_factor} = 1.0 + 0.30 \cdot \ln(\bar{d} + 1) \quad (9)$$

### 8.4 Regime-Stratified Multipliers

GRAPHPROFILER [12] classifies KGs into three structural regimes. The regime-specific multipliers  $C^{(\rho)}$  are derived from validated results on MetaQA (hub-homogeneous) and estimated for typed-heterogeneous graphs:

The hub-homogeneous constant  $C_{\text{trb}} = 11.3$  is derived directly from MetaQA: optimal  $\text{trb\_factor} \approx 26$ ,  $|R| = 9$ ,  $26/\ln(10) \approx 11.3$ .



Table 4: Regime-specific initialization multipliers. Hub-homogeneous validated on MetaQA; typed-heterogeneous estimated (cross-KB validation on Hetionet in progress).

| Regime              | $C_{\text{trb}}$ | $C_\gamma$ | $C_{r2}$ |
|---------------------|------------------|------------|----------|
| hub_homogeneous     | 11.3             | 0.80       | 2.0      |
| typed_heterogeneous | 7.0              | 1.40       | 1.5      |
| mixed               | 9.0              | 1.10       | 1.8      |

Table 5: PARAMINIT parameter summary. Universal constants are graph-invariant. Graph-relative parameters scale with measurable graph statistics.

| Parameter          | Type      | Formula                            |
|--------------------|-----------|------------------------------------|
| <b>beam_width</b>  | Universal | 10                                 |
| $\beta_s$          | Universal | 1.0                                |
| $\lambda_b$        | Universal | 0.17                               |
| <b>idf_weight</b>  | Relative  | $0.01 \cdot cv_d$                  |
| <b>vote_weight</b> | Relative  | $0.72 + 0.15Q$                     |
| <b>trb_factor</b>  | Regime    | $C \cdot \ln( R +1)$               |
| $\gamma_s$         | Regime    | $C \cdot fo_{\text{max}}/fo$       |
| <b>r2_boost</b>    | Regime    | $1.5 + C \cdot \ln(\bar{d}/ R +1)$ |
| <b>fhrb_factor</b> | Regime    | $1.0 + 0.30 \ln(\bar{d}+1)$        |

## 8.5 Summary: From Tuning to Theory

## 9 Discussion

**The hallucination-free guarantee.** All results in this paper are produced by deterministic graph traversal. Every answer is backed by an explicit, auditable edge-chain. This is a categorical property: CEREBRUM *cannot* produce an answer not reachable via graph edges. Supervised and LLM-augmented systems offer no such guarantee; their scores reflect learned probability distributions, not structural reachability.

**What the H@1 gap means.** The 40-percentage-point gap between CEREBRUM H@1 (58.9%) and UniKGQA H@1 (99.1%) is not a retrieval failure. CEREBRUM H@10 is 88.3% — meaning the correct answer is in the top 10 candidates for 88 out of 100 questions. The gap is a *ranking* problem: the system finds the answer region reliably; it does not yet consistently rank the correct answer first. Supervised methods solve this by learning from labeled examples. PARAMINIT addresses the easier portion of this gap by giving the scoring function principled starting values that do not inadvertently misrank correct answers due to arbitrary parameter initialization.

**Limitations.** PARAMINIT multipliers for typed-heterogeneous graphs are currently estimated rather

than validated. Cross-KB validation on Hetionet (47,031 entities,  $\sim 100$  relation types, **typed\_heterogeneous** regime) is in progress. The Naïve Bayes derivation of  $\lambda_b^*$  assumes path independence, which beam traversal does not guarantee; the derived constant is therefore an approximation.

## 10 Conclusion

We have presented SDRB and PARAMINIT, two contributions that together shift the treatment of scoring hyperparameters for training-free KG beam reasoning from black-box tuning targets to analytically derived quantities. SDRB replaces KB-specific per-relation boost flags with a formula derived from the graph’s own fan-out statistics, maintaining 58.9% H@1 on MetaQA 3-hop (a new training-free best) while eliminating all domain-specific parameters. SRD removes the last KB-specific component (keyword-based relation detection) with an embedding-based approach requiring no configuration on new KBs. Zipf normalization in ANOVA analysis reveals that this simplification is not a sacrifice but an uncovering: **branch\_bonus**, the first-hop ambiguity signal in beam-traversal scoring, was being suppressed by per-relation overrides. Removing the overrides exposes it, and the result is a more interpretable parameter space with a stronger theoretical foundation.

PARAMINIT provides the framework for extending this principle to all nine scoring parameters, grounding each in an established statistical technique. The goal is a system where a practitioner loads any KG, receives competitive starting parameters derived from the graph’s own statistics, and treats further tuning as optional refinement rather than a deployment prerequisite.

Code, benchmark harnesses, and tuner are available at <https://github.com/BrutalByte/CEREBRUM>.

## Acknowledgments

The author gratefully acknowledges the use of Claude (Anthropic) as a research assistant for mathematical formalization, code generation, and manuscript preparation. All conceptual contributions, architectural decisions, experimental design, and intellectual claims are solely the author’s.

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